

随机过程导论 Assignment 02

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Due Date: May 24, 2024

Problem 1

Suppose $\{X_n : n = 0, 1, \dots\}$ is a two state Markov chain

whose transition probability matrix is given by $P = \begin{bmatrix} \alpha & 1 - \alpha \\ 1 - \beta & \beta \end{bmatrix}$

Let $Z_n = (X_{n-1}, X_n)$ be a vector with four states $(0, 0), (0, 1), (1, 0), (1, 1)$

Is Z_n a Markov chain?

If yes, determine its transition probability matrix.

Solution:

我们记 $\{X_n : n = 0, 1, \dots\}$ 的两个状态为 0, 1

- Z_n 是一个 Markov 链.

我们记 $z_n = (x_{n-1}, x_n)$ ($n = 1, 2, \dots$)

其中 x_{n-1}, x_n 只能在 $\{0, 1\}$ 中取值.

则对于任意 $n = 1, 2, \dots$ 我们都有:

$$\begin{aligned} P\{Z_{n+1} = z_{n+1} | Z_n = z_n, \dots, Z_1 = z_1\} &= P\{X_{n+1} = x_{n+1} | X_n = x_n, \dots, X_1 = x_1, X_0 = x_0\} \\ &= P\{X_{n+1} = x_{n+1} | X_n = x_n, X_{n-1} = x_{n-1}\} \\ &= P\{Z_{n+1} = z_{n+1} | Z_n = z_n\} \end{aligned}$$

说明 $\{Z_n = (X_{n-1}, X_n) : n = 1, 2, \dots\}$ 具有 Markov 性, 它是一个 Markov 链.

- 下面确定 $\{Z_n = (X_{n-1}, X_n) : n = 1, 2, \dots\}$ 的状态转移矩阵:

$$\begin{aligned} P\{Z_{n+1} = z_{n+1} | Z_n = z_n\} &= P\{X_{n+1} = x_{n+1} | X_n = x_n, X_{n-1} = x_{n-1}\} \\ &= P\{X_{n+1} = x_{n+1} | X_n = x_n\} \\ &= P_{x_n, x_{n+1}} \end{aligned}$$

$$\text{因此 } \begin{cases} P\{Z_{n+1} = (0, 0) | Z_n = (0, 0)\} = P\{Z_{n+1} = (0, 0) | Z_n = (1, 0)\} = P_{0,0} = \alpha \\ P\{Z_{n+1} = (0, 1) | Z_n = (0, 0)\} = P\{Z_{n+1} = (0, 1) | Z_n = (1, 0)\} = P_{0,1} = 1 - \alpha \\ P\{Z_{n+1} = (1, 0) | Z_n = (0, 1)\} = P\{Z_{n+1} = (1, 0) | Z_n = (1, 1)\} = P_{1,0} = 1 - \beta \\ P\{Z_{n+1} = (1, 1) | Z_n = (0, 1)\} = P\{Z_{n+1} = (1, 1) | Z_n = (1, 1)\} = P_{1,1} = \beta \end{cases}$$

于是 $\{Z_n = (X_{n-1}, X_n) : n = 1, 2, \dots\}$ 的状态转移矩阵为:

$$Q = \begin{bmatrix} \alpha & 1 - \alpha & & \\ & 1 - \beta & \beta & \\ \alpha & 1 - \alpha & & \\ & 1 - \beta & \beta & \end{bmatrix}$$

Problem 2

A Markov chain with state space $\{0, 1, 2, 3\}$ has transition probability matrix

$$P = \begin{bmatrix} \frac{1}{2} & \frac{1}{4} & & \frac{1}{4} \\ & 1 & & \\ & \frac{1}{4} & \frac{1}{2} & \frac{1}{4} \\ & & & 1 \end{bmatrix}$$

(a) Draw the transition diagram.

• **Solution:**

(b) Suppose the initial probability distribution is uniform, i.e. $P\{X_0 = i\} = \frac{1}{4}$ ($i = 0, 1, 2, 3$)
Determine $P\{X_n = 1\}$ and $P\{X_n = 2\}$

• **Solution:**

$$P^{(2)} = P^2 = \begin{bmatrix} \frac{1}{4} & \frac{3}{8} & \frac{3}{8} \\ & 1 & \\ & \frac{3}{8} & \frac{1}{4} & \frac{3}{8} \\ & & & 1 \end{bmatrix}$$

$$P^{(3)} = P^3 = \begin{bmatrix} \frac{1}{8} & \frac{7}{16} & \frac{7}{16} \\ & 1 & \\ & \frac{7}{16} & \frac{1}{8} & \frac{7}{16} \\ & & & 1 \end{bmatrix}$$

$$\text{归纳可知 } P^{(n)} = P^n = \begin{bmatrix} \frac{1}{2^n} & \frac{2^n-1}{2^{n+1}} & \frac{2^n-1}{2^{n+1}} \\ & 1 & \\ & \frac{2^n-1}{2^{n+1}} & \frac{1}{2^n} & \frac{2^n-1}{2^{n+1}} \\ & & & 1 \end{bmatrix}$$

(根据 (a) 中的图像可知这个归纳结果是合理的)

$$\begin{aligned} P\{X_n = 1\} &= \sum_{i=1}^4 P\{X_n = 1|X_0 = i\}P\{X_0 = i\} \\ &= \sum_{i=1}^4 P_{i,1}^{(n)} \cdot \frac{1}{4} \\ &= \frac{1}{4} \left(\frac{2^n-1}{2^{n+1}} + 1 + \frac{2^n-1}{2^{n+1}} + 0 \right) \\ &= \frac{1}{2} - \frac{1}{2^{n+2}} \\ P\{X_n = 2\} &= \sum_{i=1}^4 P\{X_n = 2|X_0 = i\}P\{X_0 = i\} \\ &= \sum_{i=1}^4 P_{i,2}^{(n)} \cdot \frac{1}{4} \\ &= \frac{1}{4} (0 + 0 + \frac{1}{2^n} + 0) \\ &= \frac{1}{2^{n+2}} \end{aligned}$$

Problem 3

Consider independent tosses of a fair die.

Let X_n be the maximum of numbers appearing in the first n throws, $n = 1, 2, \dots$

(a) Verify that $\{X_n : n \geq 1\}$ is a Markov chain.

Describe the state space and the transition probability matrix of this homogeneous Markov Chain.

• **Solution:**

记第 n 次投掷骰子的结果为 ξ_n , 其分布为 $P\{\xi_n = i\} = \frac{1}{6}$ ($i = 1, 2, 3, 4, 5, 6$)

则对于任意 $n \in \mathbb{N}$ 都有 $X_{n+1} = \max\{\xi_1, \dots, \xi_n, \xi_{n+1}\} = \max\{X_n, \xi_{n+1}\}$

显然有 $P\{X_{n+1} = x_{n+1} | X_n = x_n, \dots, X_1 = x_1\} = P\{X_{n+1} = x_{n+1} | X_n = x_n\}$ 成立

说明 $\{X_n : n \geq 1\}$ 具有 Markov 性, 因此是一个 Markov 链.

其状态空间为 $\{1, 2, 3, 4, 5, 6\}$

$$\text{状态转移矩阵为 } P = \begin{bmatrix} \frac{1}{6} & \frac{1}{6} & \frac{1}{6} & \frac{1}{6} & \frac{1}{6} & \frac{1}{6} \\ & \frac{2}{6} & \frac{1}{6} & \frac{1}{6} & \frac{1}{6} & \frac{1}{6} \\ & & \frac{3}{6} & \frac{1}{6} & \frac{1}{6} & \frac{1}{6} \\ & & & \frac{4}{6} & \frac{1}{6} & \frac{1}{6} \\ & & & & \frac{5}{6} & \frac{1}{6} \\ & & & & & 1 \end{bmatrix}$$

(b) Find out the classes of all states.

Is there any state closed (a state is closed if, once entered, it cannot be left)?

Is there any state recurrent/transient?

Is there any state positive recurrent?

Write your reasons briefly.

Solution:

- 状态 6 是闭的 (即吸收态), 因为 $P\{X_{n+1} = 6 | X_n = 6\} = P_{6,6} = 1$
- 状态 6 是常返态 (具体来说, 是正常返态), 因为它是一个吸收态.
状态 1, 2, 3, 4, 5 是瞬时态,
因为它们都可到达状态 6, 可是一旦过程进入状态 6, 它就无法返回到其他状态.

Problem 4

Assume a Markov chain X_n has state $0, 1, 2, \dots$

and transition probabilities:
$$P_{i,0} = \frac{1}{2i^\alpha}$$

and transition probabilities:
$$P_{i,i+1} = \begin{cases} 1 & i = 0 \\ 1 - \frac{1}{2i^\alpha} & i = 1, 2, \dots \end{cases}$$
 where $\alpha > 0$ is a constant.

(a) Is this Markov chain irreducible?

• **Solution:**

- 一方面, 任意状态 $i \geq 1$ 都能到达状态 0;
- 另一方面, 从状态 0 可以到达状态 1, 从状态 1 可以到达状态 2, 以此类推...
所以从状态 0 可以到达任意状态 $i \geq 1$

因此 $\{X_n : n \in \mathbb{N}\}$ 的所有状态之间都是互通的, 因而是不可约的.

(b) Assume $X_0 = 0$ and let T be the first return time to 0

(i.e. the first time after the initial time the chain is back at the origin)

Determine for which $\alpha > 0$ is $1 - f_0 = P\{\text{no return}\} = \lim_{n \rightarrow \infty} P\{T > n\} = 0$

• **Solution:**

$$\begin{aligned} 1 - f_0 &= P\{\text{no return}\} \\ &= \lim_{n \rightarrow \infty} P\{T > n\} \\ &= \lim_{n \rightarrow \infty} P\{X_n = n | X_0 = 0\} \\ &= \lim_{n \rightarrow \infty} \left(\prod_{i=0}^{n-1} P_{i,i+1} \right) \\ &= \lim_{n \rightarrow \infty} \left\{ 1 \cdot \prod_{i=1}^{n-1} \left(1 - \frac{1}{2i^\alpha} \right) \right\} \end{aligned}$$

要寻找 $\alpha > 0$ 使得 $1 - f_0 = \lim_{n \rightarrow \infty} \prod_{i=1}^{n-1} (1 - \frac{1}{2i^\alpha}) = 0$,

等价于寻找 $\alpha > 0$ 使得 $\lim_{n \rightarrow \infty} \log\{\prod_{i=1}^{n-1} (1 - \frac{1}{2i^\alpha})\} = \lim_{n \rightarrow \infty} \sum_{i=1}^{n-1} \log(1 - \frac{1}{2i^\alpha}) = -\infty$

根据 $\log(1 - x) \approx -x$ ($x \rightarrow 0$) 可知,

等价于寻找 $\alpha > 0$ 使得 $\lim_{n \rightarrow \infty} \sum_{i=1}^{n-1} (-\frac{1}{2i^\alpha}) = -\infty$

即等价于寻找 $\alpha > 0$ 使得 $\lim_{n \rightarrow \infty} \sum_{i=1}^{n-1} \frac{1}{i^\alpha} = \sum_{i=1}^{\infty} \frac{1}{i^\alpha} = \infty$

我们知道级数 $\sum_{i=1}^{\infty} \frac{1}{i^\alpha}$ 在 $0 < \alpha \leq 1$ 时发散, 在 $\alpha > 1$ 时收敛,

因此当且仅当 $0 < \alpha \leq 1$ 时我们有 $1 - f_0 = P\{\text{no return}\} = \lim_{n \rightarrow \infty} P\{T > n\} = 0$ 成立.

(c) Depending on α , determine which classes are recurrent.

• **Solution:**

根据 (b) 的结论, 我们知道:

状态 0 在 $0 < \alpha \leq 1$ 时是常返态, 在 $\alpha > 1$ 时是瞬时态.

由于所有状态都是互通的, 因此它们要么都是常返态, 要么都是瞬时态.

因此所有状态在 $0 < \alpha \leq 1$ 时是常返态, 在 $\alpha > 1$ 时是瞬时态.

Problem 5

Let $\{X_n : n = 0, 1, 2, \dots\}$ be a simple random walk.

Each step it has equal probability to walk to the left and to the right.

The first passage time $\tau(m) = \min\{n \geq 0 : X_n = m\}$ is the first time the process ever walks to position m

Suppose that $X_0 = 0$

Denote $G_m(z) = E[z^{\tau(m)}]$ as the probability generating function of $\tau(m)$

Without loss of generality, assume that $m > 0$

(a) Show that $G_m(z) = (G_1(z))^m$

• **Lemma: (Catalan 数)**

将 n 个 -1 和 $n+1$ 个 $+1$ 排成一列,

使得任意前 i 项和 ($i = 1, 2, \dots, 2n$) 都小于 1 的排列总数是 **Catalan 数** $C_n = \frac{1}{n+1} \binom{2n}{n}$

其递推公式为 $C_n = \sum_{i=0}^{n-1} C_i \cdot C_{n-1-i}$

• **Solution:**

记 $\tau(i, j) = \min\{n \geq 0 : X_n = j, \text{ given that } X_0 = i\}$

记 $s(i, j) = \sum_{k=i}^{j-1} \tau(k, k+1)$

◦ **首先证明** $\tau(i, i+1) \stackrel{d}{=} \tau(j, j+1)$ ($\forall i, j \in \mathbb{Z}$):

任意给定 $i \in \mathbb{Z}$, 我们注意到 $\tau(i, i+1)$ 只可能取奇数值 $2n+1$ ($n \geq 0$)

对于任意 $n \geq 0$ 都有:

$$\begin{aligned} P\{\tau(i, i+1) = 2n+1\} &= P\{X_{2n+1} = 1, X_k \leq 0 \text{ for all } k = 1, 2, \dots, 2n | X_0 = 0\} \\ &= C_n \left(\frac{1}{2}\right)^n \left(\frac{1}{2}\right)^{n+1} \\ &= C_n \left(\frac{1}{2}\right)^{2n+1} \end{aligned}$$

根据 i 的任意性可知 $\tau(i, i+1) \stackrel{d}{=} \tau(j, j+1) \ (\forall i, j \in \mathbb{Z})$

我们注意到, 从状态 0 出发的过程要到达状态 $m > 0$,

必然要依次到达状态 $1, \dots, m-1$

因此我们有:

$$\begin{aligned}\tau(0, m) &= \min\{n \geq 0 : X_n = m, \text{ given that } X_0 = 0\} \\ &= \sum_{i=0}^{m-1} \min\{n \geq 0 : X_{s(0,i)+n} = j, \text{ given that } X_{s(0,i)} = i\} \\ &= \sum_{i=0}^{m-1} \min\{n \geq 0 : X_n = j, \text{ given that } X_0 = i\} \quad (\text{利用 Markov 性}) \\ &= \tau(0, 1) + \tau(1, 2) + \dots + \tau(m-1, m)\end{aligned}$$

根据 Markov 性我们还知道 $\tau(0, 1), \dots, \tau(m-1, m)$ 是相互独立的.

因此我们有 $\tau(m) = \tau(0, m) \stackrel{d}{=} \sum_{i=1}^m \xi_i$,

其中 $\{\xi_i\}$ 独立同分布, $\xi_i \stackrel{d}{=} \tau(0, 1) = \tau(1)$

因此对于任意 z 我们都有:

$$\begin{aligned}G_m(z) &= \mathbb{E}[z^{\tau(m)}] \\ &= \mathbb{E}[z^{\sum_{i=1}^m \xi_i}] \\ &= \prod_{i=1}^m \mathbb{E}[z^{\xi_i}] \\ &= \prod_{i=1}^m \mathbb{E}[z^{\tau(1)}] \\ &= \prod_{i=1}^m G_1(z) \\ &= (G_1(z))^m\end{aligned}$$

(b) Use the result of part (a), or otherwise, to find $G_m(z)$

• **Solution:**

任意给定 z 我们都有: $z^{\tau(1)} = \begin{cases} z & \text{if } X_1 = 1 \\ z^{1+\tau(2)} & \text{if } X_1 = -1 \end{cases}$

因此我们有:

$$\begin{aligned}G_1(z) &= \mathbb{E}[z^{\tau(1)}] \\ &= z \cdot \frac{1}{2} + z \cdot \mathbb{E}[z^{\tau(2)}] \cdot \frac{1}{2} \\ &= \frac{z}{2} + \frac{z}{2} G_2(z) \\ &= \frac{z}{2} + \frac{z}{2} (G_1(z))^2\end{aligned}$$

求解 $G_1(z) = \frac{z}{2} + \frac{z}{2} (G_1(z))^2$ 可得 $G_1(z) = \frac{1}{z} \pm \sqrt{\frac{1}{z^2} - 1}$

由于 $G_1(z)$ 是一个生成函数, 故对于任意 $z \in (0, 1)$ 应有 $G_1(z) < 1$ 成立.

因此我们取 $G_1(z) = \frac{1}{z} - \sqrt{\frac{1}{z^2} - 1}$

故 $G_m(z) = (G_1(z))^m = (\frac{1}{z} - \sqrt{\frac{1}{z^2} - 1})^m$

